Calculus & Optimization

Revision Sheet

Every topic on its own page • formulas, method, worked example, traps

How to use this sheet

- **One topic per page.** Nothing splits across pages — read a page, close the book, and try to reproduce the formula box from memory.
- **Read in this order on each page:** the italic line (what the topic *is*) → the formula box → "how to approach" → the worked example → the traps.
- **The dashed "My notes" box is yours.** Add the variant that caught you out, a formula your coaching notes state differently, or the page number of a question you got wrong.
- **Traps in the orange box are the marks people actually lose.** Re-read only those the night before the exam.

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1. Limits & Continuity
2. Differentiation
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1. Limits & Continuity

Memorise six standard limits and you can do almost every limit question without L'Hopital.

KEY FORMULAS

The six to know cold

$$\lim_{x\to 0}\frac{\sin x}{x}=1, \qquad \lim_{x\to 0}\frac{\tan x}{x}=1, \qquad \lim_{x\to 0}\frac{1-\cos x}{x^2}=\tfrac{1}{2}$$

$$\lim_{x\to 0}\frac{e^x-1}{x}=1, \qquad \lim_{x\to 0}\frac{\ln(1+x)}{x}=1, \qquad \lim_{x\to\infty}\left(1+\tfrac{1}{x}\right)^x=e$$

Continuity at a

$$\lim_{x\to a} f(x)=f(a)$$

WHAT YOU MUST KNOW

- L'Hôpital applies **only** to $\frac{0}{0}$ or $\frac{\infty}{\infty}$ — check the form first.
- For a $\frac{0}{0}$ form, a **series expansion** is often faster: $\ln(1+u)\approx u-\frac{u^2}{2}$, $\cos x\approx 1-\frac{x^2}{2}$.
- Differentiable $\Rightarrow$ continuous, but **not** conversely: $|x|$ and ReLU are continuous yet not differentiable at 0.
- A **jump** discontinuity is where the two one-sided limits are finite but unequal.

HOW TO APPROACH A QUESTION

1. Substitute first — if you get a number, that is the answer.
2. If $\frac{0}{0}$: factor and cancel, or expand in series, or use a standard limit.
3. Split a product/sum of logs into separate limits: $\ln(AB)=\ln A+\ln B$.
4. For a piecewise function to be smooth, match both the *value* and the *derivative* at the join.

WORKED EXAMPLE

Q. Evaluate $\displaystyle\lim_{x\to 0}\frac{\ln\big((x^2+1)\cos x\big)}{x^2}$.

Split the log: $\dfrac{\ln(1+x^2)}{x^2}+\dfrac{\ln(\cos x)}{x^2}$. First $\to 1$. For the second, $\cos x\approx 1-\frac{x^2}{2}$ so $\ln\cos x\approx -\frac{x^2}{2}$, giving $-\frac{1}{2}$.

$$1-\tfrac{1}{2}=\mathbf{0.5}$$

WATCH OUT

- Applying L'Hôpital to a form that is not indeterminate.
- Assuming continuity implies differentiability.

MY NOTES

2. Differentiation

Standard rules, but GATE DA picks the functions from machine learning — sigmoid, softplus, tanh, ReLU.

KEY FORMULAS

Rules

$$(uv)' = u'v + uv', \qquad \left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}, \qquad \frac{d}{dx} f(g(x)) = f'(g)\, g'$$

The ML derivatives

$$\sigma' = \sigma(1-\sigma), \qquad \frac{d}{dx}\ln(1+e^x) = \sigma(x), \qquad \tanh' = 1 - \tanh^2$$

Linear approximation

$$f(x) \approx f(a) + f'(a)(x-a)$$

WHAT YOU MUST KNOW

- $\sigma' = \sigma(1-\sigma)$ means you can differentiate the sigmoid **without** differentiating — if you know σ, you know σ'.
- ReLU is not differentiable at 0 but is everywhere else (subgradient used in practice).
- $(x^n)' = nx^{n-1}$, $(e^x)' = e^x$, $(\ln x)' = \frac{1}{x}$.
- For a piecewise function, differentiability at the join requires equal one-sided derivatives.

HOW TO APPROACH A QUESTION

1. Identify the outer and inner function before applying the chain rule.
2. If the question gives you $f(x)$'s *value* rather than x, look for an identity like $\sigma' = \sigma(1-\sigma)$.
3. For piecewise-smoothness questions, write two equations (value match, slope match) per join and solve.

WORKED EXAMPLE

Q. $f(x) = \dfrac{1}{1+e^{-x}}$. Find f' at the point where $f(x) = 0.4$.

Use the identity rather than the quotient rule:

$$f' = f(1-f) = 0.4 \times 0.6 = \mathbf{0.24}$$

WATCH OUT

- Grinding the quotient rule when $\sigma' = \sigma(1-\sigma)$ gives it in one step.
- Matching only the value (not the slope) when fitting a piecewise function.

MY NOTES

3. Taylor & Maclaurin Series

Four expansions cover everything, and they double as the fastest way to kill an indeterminate limit.

KEY FORMULAS

General form ($a = 0$ gives Maclaurin)

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

The four to memorise

$$e^x = 1 + x + \tfrac{x^2}{2!} + \cdots, \qquad \sin x = x - \tfrac{x^3}{3!} + \cdots$$

$$\cos x = 1 - \tfrac{x^2}{2!} + \cdots, \qquad \ln(1+x) = x - \tfrac{x^2}{2} + \tfrac{x^3}{3} - \cdots$$

WHAT YOU MUST KNOW

- Coefficient of x^n is $\dfrac{f^{(n)}(0)}{n!}$ — so the x^3 coefficient of $\sin x$ is $-\frac{1}{3!} \approx -0.167$.
- $\dfrac{1}{1-x} = 1 + x + x^2 + \cdots$ for $|x| < 1$ (geometric series).
- First-order Taylor is the tangent line; second-order adds $\frac{1}{2}f''(a)(x-a)^2$.
- Taylor's remainder term bounds the approximation error.

HOW TO APPROACH A QUESTION

1. For a numerical estimate, substitute into the first two or three terms as instructed.
2. For a limit, expand numerator and denominator to the same order and cancel.
3. For a coefficient question, use $f^{(n)}(0)/n!$ rather than expanding fully.

WORKED EXAMPLE

Q. Estimate $e^{0.1}$ using terms up to x^2.

$$1 + 0.1 + \frac{0.1^2}{2} = 1 + 0.1 + 0.005 = \mathbf{1.1050}$$

(True value 1.10517 — two terms of error $\approx 1.7 \times 10^{-5}$.)

WATCH OUT

- Forgetting the factorials in the denominators.
- Using the $\ln(1+x)$ series outside $|x| < 1$ where it does not converge.

MY NOTES

4. Maxima, Minima & the Second-Derivative Test

The single most repeated calculus idea in GATE DA. Find where the slope vanishes, then let the second derivative classify it.

KEY FORMULAS

Stationary points

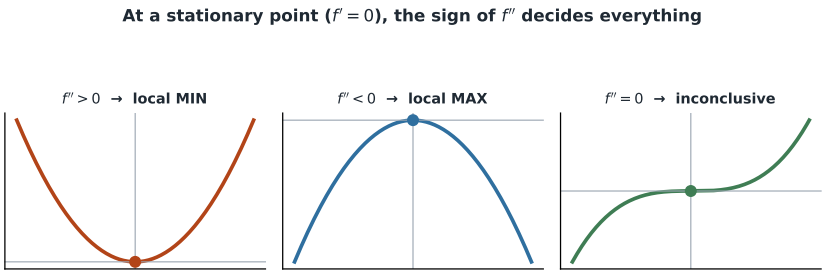
$$f'(x^*) = 0$$

Classification

$$f''(x^*) > 0 \Rightarrow \textbf{min}, \qquad f''(x^*) < 0 \Rightarrow \textbf{max}, \qquad f''(x^*) = 0 \Rightarrow \textbf{inconclusive}$$

WHAT YOU MUST KNOW

- On a **closed** interval, extrema occur at critical points *or at the endpoints* — always check both.
- An **inflection point** is where f'' changes sign, not merely where $f'' = 0$.
- "Local" is not "global": a local minimum is global only if the function is convex.
- For a cubic $ax^3 + \ldots$, f' is a quadratic — so at most two critical points.



HOW TO APPROACH A QUESTION

1. Differentiate and **factor** f' to get all critical points cleanly.
2. Evaluate f'' at each; read off min / max from the sign.
3. If the domain is an interval, also evaluate f at the endpoints and compare values.
4. For MSQ options about roots, factor f itself and locate the roots exactly.

WORKED EXAMPLE

Q. $f(x) = \frac{x^4}{4} - \frac{2x^3}{3} - \frac{3x^2}{2} + 1$. Classify its critical points.

$$f' = x^3 - 2x^2 - 3x = x(x-3)(x+1) \;\Rightarrow\; x = -1, 0, 3$$

$$f'' = 3x^2 - 4x - 3: \quad f''(0) = -3 < 0, \quad f''(3) = 12 > 0, \quad f''(-1) = 4 > 0$$

So $x = 0$ is a local **max**; $x = 3$ and $x = -1$ are local **minima**.

WATCH OUT

- Forgetting the endpoints on a closed or half-open interval.
- Declaring a maximum "the only one" without checking the endpoint value.

MY NOTES

5. Single-Variable Optimization & Convexity

Convexity is what makes optimization safe: for a convex function, a local minimum is automatically the global one.

KEY FORMULAS

Convexity test

$$f \text{ convex on an interval} \iff f''(x) \geq 0 \text{ there}$$

Gradient descent

$$x \leftarrow x - \eta\, f'(x)$$

WHAT YOU MUST KNOW

- For a convex f, any local minimum is **global**; strict convexity makes the minimiser **unique**.
- x^2 and e^x are convex on all of $\mathbb{R}$; $-x^2$ is concave; $\sin x$ is neither.
- The sum of convex functions is convex; f is concave exactly when $-f$ is convex.
- Gradient descent moves *opposite* the derivative, i.e. downhill. On $f = \frac{1}{2}w^2$ it converges for $0 < \eta < 2$.

HOW TO APPROACH A QUESTION

1. For an applied optimum, write the quantity as a function of one variable using the constraint.
2. Set the derivative to zero, solve, and confirm with the second derivative.
3. For gradient-descent questions, just evaluate f' at the current point and take one step.

WORKED EXAMPLE

Q. Minimise $f(x) = x^2 + \dfrac{16}{x}$ for $x > 0$.

$$f' = 2x - \frac{16}{x^2} = 0 \;\Rightarrow\; 2x^3 = 16 \;\Rightarrow\; x = \mathbf{2}$$

$f'' = 2 + \frac{32}{x^3} > 0$ for $x > 0$, so it is a minimum (value $f(2) = 12$).

WATCH OUT

- Solving $f' = 0$ but never checking f'' (or the domain restriction $x > 0$).
- Assuming every stationary point of a non-convex function is global.

MY NOTES

6. Multivariable & Constrained Optimization

Same idea in more dimensions: the gradient replaces the derivative, and the Hessian replaces the second derivative.

KEY FORMULAS

First-order condition

$$\nabla f = \left(\tfrac{\partial f}{\partial x}, \tfrac{\partial f}{\partial y}\right) = \mathbf{0}$$

Second-order (Hessian H)

$$H \text{ PD} \Rightarrow \min, \quad H \text{ ND} \Rightarrow \max, \quad H \text{ indefinite} \Rightarrow \text{saddle}$$

Lagrange multipliers

$$\nabla f = \lambda \nabla g \quad \text{subject to} \quad g(x, y) = 0$$

WHAT YOU MUST KNOW

- The gradient points in the direction of **steepest ascent**; $-\nabla f$ is steepest descent.
- A quadratic form $x^{\mathsf{T}} A x$ is convex $\iff$ A is positive semidefinite — the bridge to Linear Algebra.
- For a symmetric 2×2 Hessian, PD $\iff$ top-left entry > 0 and $\det > 0$.
- A saddle point satisfies $\nabla f = 0$ but is neither a max nor a min.

HOW TO APPROACH A QUESTION

1. Set both partial derivatives to zero and solve the system for the critical point.
2. Build the Hessian and test its definiteness to classify.
3. With a constraint, form $\nabla f = \lambda \nabla g$ and solve together with $g = 0$ — usually symmetric, so try $x = y$.

WORKED EXAMPLE

Q. Minimise $f = x^2 + y^2$ subject to $x + y = 1$.
$\nabla f = (2x, 2y)$, $\nabla g = (1, 1)$, so $2x = \lambda = 2y \Rightarrow x = y$. With $x + y = 1$:

$$x = y = \mathbf{0.5}, \qquad f_{\min} = 0.25 + 0.25 = \mathbf{0.5}$$

WATCH OUT

- Forgetting to substitute back into the constraint to find λ or the actual minimum value.
- Concluding "minimum" from $\nabla f = 0$ alone without checking the Hessian.

MY NOTES